

## RUNNING THE PROGRAMME FOR THREE CIRCUITS TO DEMONSTRATE EACH SCENARIO CLEARLY

### Test Case 1: Standard Valid Circuit (Design Accepted)

This proves your Ohm's Law and power formulas are working perfectly.

- **LED Name/Colour:** Red Indicator LED
- **Supply Voltage (V):** 5
- **LED Forward Voltage (V):** 2
- **Desired LED Current (mA):** 20

#### Expected output

```
-Recommended Resistor: 150 ohms  
-Minimum Resistor Power: 0.06 W  
-Design Status: Design Accepted
```

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### Test Case 2: Out of Bounds Current (High Current Warning)

This proves your safety classification rules work.

- **LED Name/Colour:** High Power White LED
- **Supply Voltage (V):** 12
- **LED Forward Voltage (V):** 3.4
- **Desired LED Current (mA):** 40 (This is > 30mA )

#### Expected output

```
Recommended Resistor: 215 ohms  
Minimum Resistor Power: 0.344 W  
Design Status: Warning: LED current may be too high for a  
standard LED.
```

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### Test Case 3: Invalid Design (Input Validation Check)

This proves your system actively rejects wrong electrical inputs and prevents system crashes.

- **LED Name/Colour:** Blue LED
- **Supply Voltage (V):** 3
- **LED Forward Voltage (V):** 3.2 (Supply is less than Forward Voltage )
- **Desired LED Current (mA):** 20

#### Expected Output

```
-Design Status: Invalid Design
```

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